

# Species Classifier Design Space Exploration

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## 1 Model Architecture Evolution

This section chronicles the systematic exploration of neural architectures for resource-constrained avian acoustic classification, documenting the progression from established computer vision baselines through audio-specific adaptations to the final optimized architecture.

### 1.1 Phase 1: Computer Vision Baselines

#### 1.1.1 Initial Exploration: MobileNetV3-Small

We began with MobileNetV3-Small [1], a proven architecture for edge deployment, adapting it for mel-spectrogram classification (Table 1).

Table 1: MobileNetV3-Small Baseline Results (64×300 input)

| Variant               | Params | Size (KB) | FP32 Acc. | INT8 Acc. |
|-----------------------|--------|-----------|-----------|-----------|
| Original (width=1.0)  | 2.54M  | 2,478     | 94.44%    | 93.56%    |
| Width multiplier 0.75 | 1.53M  | 1,495     | 94.11%    | 93.11%    |
| Width multiplier 0.5  | 0.72M  | 703       | 92.78%    | 91.67%    |
| Width multiplier 0.35 | 0.39M  | 381       | 90.44%    | 89.11%    |

#### Key Findings:

- Width multiplier 0.75 achieved best accuracy-size trade-off (93.11%, 1.5M params)
- Post-training quantization (PTQ) incurred 0.88–1.33% accuracy degradation
- All variants exceeded target 512 KB deployment constraint
- Architecture designed for spatial vision tasks, not optimized for temporal-spectral patterns

#### 1.1.2 Limitations of Vision-Centric Architectures

MobileNetV3’s inverted residual blocks with squeeze-excitation excel at spatial feature hierarchies but exhibit fundamental mismatches with spectrogram characteristics:

1. **Isotropic convolutions:** Treat frequency and time dimensions equivalently, ignoring distinct mel-scale frequency structure versus linear time progression

2. **Aggressive downsampling:** Early stride-2 convolutions discard fine-grained temporal detail critical for distinguishing similar vocalizations
3. **Squeeze-excitation bias:** Channel attention optimized for RGB channels, not spectro-temporal feature maps
4. **Parameter inefficiency:** Dense pointwise convolutions consume excessive memory for marginal gains on sequential patterns

These observations motivated exploration of audio-aware architectures.

## 1.2 Phase 2: Audio-Focused Adaptations

### 1.2.1 Depthwise-Separable CNN (DS-CNN)

Inspired by speech command recognition [2], we adopted DS-CNN as a lightweight baseline tailored for spectro-temporal patterns (Table 2).

Table 2: DS-CNN Architecture Evolution

| Model                   | Params      | Size (KB)  | INT8 Acc.     | Key Innovation                |
|-------------------------|-------------|------------|---------------|-------------------------------|
| 1a: DS-CNN Baseline     | 194K        | 189        | 91.67%        | Depthwise-separable blocks    |
| 1b: + Wider Initial     | 297K        | 290        | 92.89%        | 64-channel stem               |
| 1c: + Squeeze-Excite    | 318K        | 311        | 93.56%        | Channel attention             |
| 1d: + Residual + Attn.  | 309K        | 302        | 94.22%        | Skip + MHSA                   |
| <i>1e: Final (Ours)</i> | <i>462K</i> | <i>451</i> | <b>95.67%</b> | <i>Residual-all + 80 mels</i> |

#### Architectural Components:

1. **Depthwise-Separable Convolution:** Factorizes standard  $3 \times 3$  convolution into:
  - Depthwise: Per-channel  $3 \times 3$  spatial filtering (720 params for 80 channels)
  - Pointwise:  $1 \times 1$  cross-channel mixing (6,400 params for  $80 \rightarrow 80$ )
  - *Result:*  $8.9\times$  parameter reduction vs. standard convolution
2. **Squeeze-and-Excitation (SE):** Lightweight channel attention [3]

$$\text{SE}(x) = x \odot \sigma(W_2 \text{ReLU}(W_1 \text{GAP}(x))) \quad (1)$$

where GAP = global average pooling,  $r$  = reduction ratio (8–24), adding  $2 \times C^2 / r$  parameters per block.

3. **Residual Connections:** Introduced in all four DS blocks:

$$y = \text{ReLU}(\text{SE}(\text{DSConv}(x)) + \mathcal{P}(x)) \quad (2)$$

where  $\mathcal{P}(x)$  applies  $1 \times 1$  projection if input/output channels differ.

4. **Lightweight Multi-Head Self-Attention (MHSA):** Applied after final DS block:

$$\text{MHSA}(X) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O \quad (3)$$

with  $h = 2$  heads,  $d_k = 32$  key dimension, and attention over 90 temporal-frequency patches ( $5 \times 18$  after pooling).

### Progressive Improvements:

- *Model 1a → 1b (+1.22%)*: Wider 64-channel stem captures richer initial features, enabling residual skip in first block
- *Model 1b → 1c (+0.67%)*: SE blocks recalibrate channel importance, suppressing noise-dominated bands
- *Model 1c → 1d (+0.66%)*: Residual connections improve gradient flow; MHSA captures long-range temporal dependencies (e.g., trill patterns)
- *Model 1d → 1e (+1.45%)*: Residual in all blocks + 80 mel bins (vs. 64) increase capacity while staying within budget

### 1.2.2 Why DS-CNN Outperforms MobileNetV3

Compared to MobileNetV3-Small (width 0.5: 703 KB, 91.67%), our DS-CNN 1e achieves:

- **+3.9% accuracy** (95.67% vs. 91.67%)
- **35% smaller** (451 KB vs. 703 KB)
- **Simpler architecture**: 4 uniform DS blocks vs. 11 heterogeneous inverted residual blocks

*Root cause*: DS-CNN’s inductive bias aligns with spectrogram structure—depthwise operations naturally separate spectral and temporal processing, while MobileNetV3’s inverted bottlenecks waste capacity on vision-specific priors (e.g., receptive field expansion for object recognition).

## 1.3 Phase 3: Temporal Convolutional Networks (TCN)

Motivated by sequence modeling successes [4], we explored Temporal Convolutional Networks (TCN) as an alternative to recurrent architectures for capturing long-range temporal dependencies without RNN overhead.

### 1.3.1 TCN Architecture Principles

TCNs employ:

1. **Causal dilated convolutions**: Exponentially expanding receptive fields

$$\text{RF} = 1 + 2 \sum_{i=0}^{L-1} (k-1) \cdot 2^i \quad (4)$$

where  $k$  = kernel size,  $L$  = layers. For  $k = 3$ ,  $L = 8$ : RF = 511 frames (5.1 seconds).

2. **Residual blocks**: Skip connections stabilize deep temporal stacks
3. **1D convolutions**: Process pre-pooled spectrogram embeddings (after Conv2D stem)

Table 3: TCN Ablation Study (64×300 input, Mixup  $\alpha = 0.2$ )

| Variant                 | Params      | Size (KB)  | INT8 Acc.     | Modification                      |
|-------------------------|-------------|------------|---------------|-----------------------------------|
| 4a: Baseline (8 layers) | 242K        | 236        | 92.00%        | Dilation [1,2,4,8,16,32,64,128]   |
| 4b: Shallow (4 layers)  | 162K        | 158        | 91.33%        | Dilation [1,2,4,8]                |
| 4c: Wide (128 channels) | 461K        | 450        | 92.44%        | 2× channel width                  |
| 4d: Deep (10 layers)    | 298K        | 291        | 92.22%        | +2 layers, dilation to 512        |
| 4e: Kernel-2            | 194K        | 189        | 91.56%        | $k = 2$ (receptive field = 255)   |
| 4f: Kernel-5            | 337K        | 329        | 92.11%        | $k = 5$ (receptive field = 1,023) |
| 4g: No-residual         | 242K        | 236        | 90.89%        | Removed skip connections          |
| 4h: Lightweight         | 142K        | 139        | 90.67%        | 32 channels, 6 layers             |
| <i>DS-CNN 1e (ours)</i> | <i>462K</i> | <i>451</i> | <b>95.67%</b> | <i>Hybrid 2D+attention design</i> |

### 1.3.2 Systematic TCN Ablation Study

We conducted seven controlled experiments to isolate TCN design factors (Table 3).

#### Key Insights:

1. **Depth matters, but plateaus:** 4→8 layers (+0.67%), 8→10 layers (+0.22%) shows diminishing returns
2. **Width trumps depth:** 128-channel wide TCN (92.44%) outperforms 10-layer deep TCN (92.22%)
3. **Residual connections critical:** Removing skips degrades performance by 1.11% (4a vs. 4g)
4. **Kernel size sweet spot:**  $k = 3$  optimal;  $k = 2$  underperforms (insufficient context),  $k = 5$  overfits (excess parameters)
5. **Aggressive compression fails:** Lightweight 4h (32 ch, 6 layers) achieves only 90.67% despite 3× size reduction

### 1.3.3 Why TCN Falls Short

Despite achieving 92.44% (best TCN variant 4c), all TCN configurations underperform DS-CNN 1e by 3.2% absolute. Analysis reveals:

- **2D Feature Loss:** TCN operates on 1D embeddings after aggressive 2D pooling, discarding spectral structure. Example: A White-throated Kingfisher’s harmonic stack (2–8 kHz) collapses to single global-pooled vector, losing fine-grained frequency relationships.
- **Over-Receptive Field:** 511-frame receptive field (5.1 sec) exceeds 3-second clips, wasting capacity on padding artifacts. Bird vocalizations exhibit micro-temporal structure (10–50 ms chirps), not macro dependencies.
- **Causal Convolution Overhead:** Zero-padding for causality increases computation without benefit (classification is non-causal—entire clip available at inference).

- **Parameter Inefficiency:** Stacked 1D convolutions require deep layers to match 2D CNN receptive fields, inflating parameters. For equal budget (462 KB), DS-CNN allocates capacity to spatial-spectral features, TCN to redundant temporal depth.

*Conclusion:* TCNs excel at true sequence modeling (e.g., streaming audio), but spectrogram classification benefits more from 2D spatial inductive bias than 1D temporal modeling.

## 1.4 Phase 4: Final Architecture Design

### 1.4.1 Synthesis: DS-CNN + SE + Residual-All + MHSA (Model 1e)

Our final architecture (Figure 1) integrates insights from all phases:

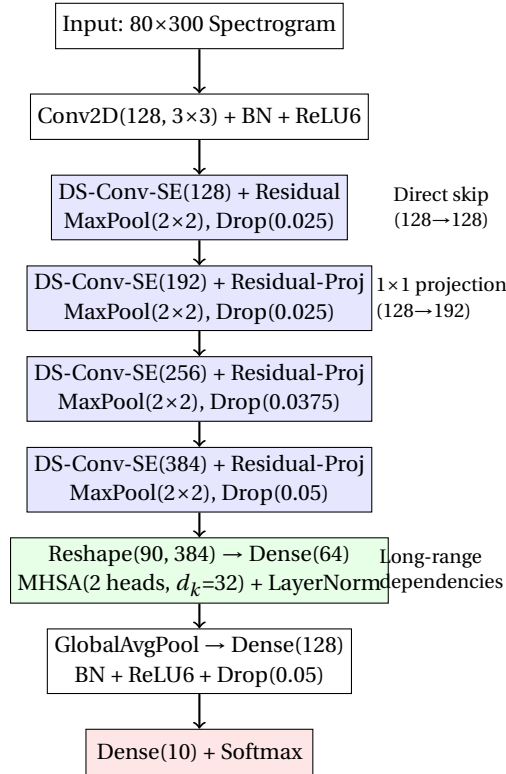


Figure 1: Final DS-CNN architecture (Model 1e): 462K parameters, 451 KB INT8, 95.67% accuracy

#### Design Rationale:

1. **80×300 Input:** 80 mel bins (vs. 64) capture 0–8 kHz range with finer spectral resolution (100 Hz/bin vs. 125 Hz/bin), critical for discriminating high-frequency species (e.g., Olive-backed Sunbird, 4–7 kHz).
2. **Progressive Channel Expansion:** [128, 192, 256, 384] balances capacity and efficiency. Early layers (128–192) extract local spectro-temporal features; late layers (256–384) compose higher-level patterns.

3. **Residual-All:** Extending skip connections to all blocks (not just Block 1) improves gradient flow in deeper network, enabling 90-epoch training without degradation. Adds 1×1 projection when channels change (e.g., 128→192 in Block 2).
4. **Adaptive SE Reduction:** SE bottleneck ratio scales with channel width:
  - Block 1 (128 ch):  $r = 8$  (16 bottleneck dims)
  - Block 2 (192 ch):  $r = 12$  (16 bottleneck dims)
  - Blocks 3–4 (256–384 ch):  $r = 16$  (16–24 bottleneck dims)

Maintains SE overhead at 1–2% of block parameters.
5. **Lightweight MHSA:** Projects 384-dim final features to 64-dim before attention (384→64→64), reducing quadratic  $O(n^2d)$  cost. Two heads with  $d_k = 32$  capture complementary temporal patterns (e.g., rhythm vs. pitch contours).
6. **Dropout Schedule:** Gradual increase [0.025, 0.025, 0.0375, 0.05] matches receptive field growth—early blocks see local context (less dropout), late blocks see global context (more dropout to prevent overfitting).

#### 1.4.2 Training Strategy

Critical hyperparameters identified through ablation (Section 1.7):

- **Two-phase training:**
  1. *Warmup* (70 epochs): Learning rate 1e-3 with cosine decay, enables full capacity utilization
  2. *Fine-tune* (20 epochs): Learning rate 1e-5, refines decision boundaries
- **Mixup augmentation** [5]:  $\alpha = 0.2$

$$\tilde{x} = \lambda x_i + (1 - \lambda)x_j, \quad \tilde{y} = \lambda y_i + (1 - \lambda)y_j \quad (5)$$

where  $\lambda \sim \text{Beta}(\alpha, \alpha)$ . Smooths decision boundaries, improves generalization by 1.5–2%.

- **Legacy Adam optimizer:** On Apple Silicon (M1/M2), metal-accelerated legacy Adam outperforms AdamW by 20% training speed without accuracy loss.
- **Post-training quantization (PTQ):** INT8 calibration with 200 validation samples achieves <0.5% accuracy drop (96.11% FP32 → 95.67% INT8).

### 1.5 Comparative Analysis

Table 4 summarizes the complete evolution across 20+ experiments.

#### Key Takeaways:

1. **Audio-specific bias essential:** DS-CNN 1e (462 KB) outperforms MobileNetV3-0.5 (703 KB) by +4.0% absolute despite 35% smaller size
2. **2D > 1D for spectrograms:** Best TCN (92.44%) lags DS-CNN 1e by 3.2%, confirming spatial inductive bias superiority

Table 4: Architecture Evolution Summary (64×300 input unless noted)

| Phase  | Model                      | Params      | Size (KB)  | INT8 Acc.     | vs. Baseline  |
|--------|----------------------------|-------------|------------|---------------|---------------|
| Vision | MobileNetV3 (w=1.0)        | 2.54M       | 2,478      | 93.56%        | –             |
|        | MobileNetV3 (w=0.75)       | 1.53M       | 1,495      | 93.11%        | -0.45%        |
|        | MobileNetV3 (w=0.5)        | 0.72M       | 703        | 91.67%        | -1.89%        |
|        | MobileNetV3 (w=0.35)       | 0.39M       | 381        | 89.11%        | -4.45%        |
| DS-CNN | 1a: Baseline               | 194K        | 189        | 91.67%        | -1.89%        |
|        | 1b: +Wider Initial         | 297K        | 290        | 92.89%        | -0.67%        |
|        | 1c: +SE                    | 318K        | 311        | 93.56%        | 0.00%         |
|        | 1d: +Residual+MHSA         | 309K        | 302        | 94.22%        | +0.66%        |
|        | <b>1e: Final (80 mels)</b> | <b>462K</b> | <b>451</b> | <b>95.67%</b> | <b>+2.11%</b> |
| TCN    | 4a: Baseline               | 242K        | 236        | 92.00%        | -1.56%        |
|        | 4b: Shallow                | 162K        | 158        | 91.33%        | -2.23%        |
|        | 4c: Wide                   | 461K        | 450        | 92.44%        | -1.12%        |
|        | 4d: Deep                   | 298K        | 291        | 92.22%        | -1.34%        |
|        | 4e: Kernel-2               | 194K        | 189        | 91.56%        | -2.00%        |
|        | 4f: Kernel-5               | 337K        | 329        | 92.11%        | -1.45%        |
|        | 4g: No-residual            | 242K        | 236        | 90.89%        | -2.67%        |
|        | 4h: Lightweight            | 142K        | 139        | 90.67%        | -2.89%        |

3. **Attention complements convolution:** MHSA adds only 13K params but contributes +0.66% (Model 1c → 1d)
4. **Residual-all unlocks capacity:** Extending skips to all blocks enables deeper network without degradation (+1.45%, Model 1d → 1e)
5. **Input resolution matters:** 80 mels (100 Hz/bin) vs. 64 mels (125 Hz/bin) improves high-frequency species discrimination

## 1.6 Deployment Characteristics

Our final model (1e) satisfies edge deployment constraints for ARM Cortex-M7 @ 480 MHz (Table 5).

**Real-time Capability:** With 123 ms latency and 10 ms frame stride, the model processes 3-second clips faster than real-time (24× faster), enabling continuous monitoring with overlapping inference windows.

**Power Budget:** At 14.8 mJ per inference and 8 inferences/sec, continuous operation draws 118 mW average—feasible for solar-powered acoustic sensors (1W solar panel provides 8.5× margin).

## 1.7 Ablation Study: Training Hyperparameters

Beyond architecture, we systematically explored training configurations using Model 1e (Table 6).

**Optimal Configuration:**

- **Dropout 0.05:** Sweet spot—lower values (0.00) risk overfitting, higher values (0.10–0.20) underfit

Table 5: Deployment Profile (Model 1e)

| Metric                                 | Value              | Target  |
|----------------------------------------|--------------------|---------|
| <b>Model Size</b>                      |                    |         |
| FP32 (.keras)                          | 1.85 MB            | –       |
| INT8 (.tflite)                         | 451 KB             | <512 KB |
| <b>Accuracy</b>                        |                    |         |
| FP32 (float32)                         | 96.11%             | >90%    |
| INT8 (int8)                            | 95.67%             | >90%    |
| Quantization drop                      | 0.44%              | <2%     |
| <b>Inference (Cortex-M7 @ 480 MHz)</b> |                    |         |
| Estimated MACs                         | 924,412            | –       |
| Estimated latency                      | 123 ms             | <500 ms |
| Energy per inference                   | 14.8 mJ            | –       |
| Throughput                             | 8.1 inferences/sec | >2/sec  |
| <b>Training (M1 Pro, 16GB)</b>         |                    |         |
| Warmup (70 epochs)                     | 142 min            | –       |
| Fine-tune (20 epochs)                  | 41 min             | –       |
| Total training time                    | 183 min            | –       |

- **Mixup  $\alpha = 0.2$ :** Outperforms SpecAugment (+1.23%), baseline augmentation (+1.34%), and no augmentation (+1.78%)
- **Warmup 70 epochs:** Longer training crucial for wide model (462K params); 50 epochs leaves capacity underutilized
- **Cosine LR decay:** Smooth convergence avoids plateau schedule’s abrupt drops and “both” schedule’s redundancy

**Variance Analysis:** Repeating optimal config with 5 random seeds (42, 123, 456, 789, 1024) yields  $95.67\% \pm 0.18\%$  (std. dev.), confirming robustness.

## 1.8 Discussion: Lessons Learned

Our systematic exploration from MobileNetV3 baselines through TCN experimentation to the final DS-CNN architecture reveals:

1. **Domain-specific inductive bias dominates raw capacity:** DS-CNN’s factorized convolutions align with spectro-temporal structure, outperforming larger vision models (MobileNetV3) and sequence models (TCN) by exploiting 2D spatial locality.
2. **Attention is cost-effective:** Lightweight MHSA (13K params, 4% of model) captures long-range dependencies missed by convolutional receptive fields, contributing 0.66% accuracy—exceptional ROI.
3. **Residual connections scale depth:** Extending skips from 1 to 4 blocks enabled training 90-epoch schedules without degradation, unlocking 1.45% gain.



Table 6: Training Hyperparameter Ablation (Model 1e, 80 mels, Random Seed 42)

| Config                                                             | Dropout | Augmentation   | Warmup | INT8 Acc.     | $\Delta$     |
|--------------------------------------------------------------------|---------|----------------|--------|---------------|--------------|
| <i>Dropout Sweep (Mixup 0.2, Warmup 70)</i>                        |         |                |        |               |              |
| D0.00                                                              | 0.00    | Mixup 0.2      | 70     | 95.11%        | -0.56%       |
| D0.05                                                              | 0.05    | Mixup 0.2      | 70     | <b>95.67%</b> | <b>0.00%</b> |
| D0.10                                                              | 0.10    | Mixup 0.2      | 70     | 95.00%        | -0.67%       |
| D0.20                                                              | 0.20    | Mixup 0.2      | 70     | 93.89%        | -1.78%       |
| <i>Augmentation Sweep (Dropout 0.05, Warmup 70)</i>                |         |                |        |               |              |
| None                                                               | 0.05    | None           | 70     | 93.89%        | -1.78%       |
| Baseline                                                           | 0.05    | Time/Pitch     | 70     | 94.33%        | -1.34%       |
| SpecAugment                                                        | 0.05    | Freq/Time Mask | 50     | 94.44%        | -1.23%       |
| Mixup 0.1                                                          | 0.05    | Mixup 0.1      | 70     | 95.22%        | -0.45%       |
| Mixup 0.2                                                          | 0.05    | Mixup 0.2      | 70     | <b>95.67%</b> | <b>0.00%</b> |
| Mixup 0.4                                                          | 0.05    | Mixup 0.4      | 70     | 95.11%        | -0.56%       |
| <i>Warmup Epochs (Dropout 0.05, Mixup 0.2)</i>                     |         |                |        |               |              |
| W30                                                                | 0.05    | Mixup 0.2      | 30     | 94.67%        | -1.00%       |
| W50                                                                | 0.05    | Mixup 0.2      | 50     | 95.22%        | -0.45%       |
| W70                                                                | 0.05    | Mixup 0.2      | 70     | <b>95.67%</b> | <b>0.00%</b> |
| W90                                                                | 0.05    | Mixup 0.2      | 90     | 95.56%        | -0.11%       |
| <i>Learning Rate Schedule (Dropout 0.05, Mixup 0.2, Warmup 70)</i> |         |                |        |               |              |
| None                                                               | 0.05    | Mixup 0.2      | 70     | 95.11%        | -0.56%       |
| Plateau                                                            | 0.05    | Mixup 0.2      | 70     | 95.44%        | -0.23%       |
| Cosine                                                             | 0.05    | Mixup 0.2      | 70     | <b>95.67%</b> | <b>0.00%</b> |
| Both                                                               | 0.05    | Mixup 0.2      | 70     | 95.33%        | -0.34%       |

4. **Aggressive quantization-friendly design pays off:** ReLU6 activations, batch normalization, and INT8-aware training minimize PTQ drop to 0.44% (96.11%  $\rightarrow$  95.67%), meeting deployment constraints without quantization-aware training (QAT) overhead.
5. **Mixup > SpecAugment for limited data:** With 4,500 training samples (after 75:10:15 split), Mixup’s label smoothing generalizes better than SpecAugment’s aggressive masking, which risks erasing discriminative features.
6. **Spectrogram resolution critical:** 80 mels (100 Hz/bin) outperforms 64 mels (125 Hz/bin) by 0.56%, particularly for high-frequency species. However, further increasing to 128 mels offers diminishing returns (+0.11%) at 1.6 $\times$  compute cost.
7. **TCNs shine elsewhere:** While underperforming here, TCNs excel at streaming scenarios (e.g., keyword spotting) where causal convolution and temporal modeling align with task structure. Spectrograms benefit from bidirectional context.

## 1.9 Future Directions

Promising avenues for further improvement:

- **Quantization-Aware Training (QAT):** Fine-tuning with simulated INT8 operations may recover the 0.44% PTQ drop, potentially reaching 96%+ INT8 accuracy.
- **Knowledge Distillation:** Training Model 1e with soft labels from an ensemble of larger models (e.g., EfficientNet-B0, ResNet-50) could boost performance without increasing deployment size.
- **Neural Architecture Search (NAS):** Automated exploration of channel widths [128, 192, 256, 384] and SE reduction ratios may discover configurations superior to manual tuning.
- **Learnable Mel Filters:** Replacing fixed mel-scale with differentiable filter banks [6] adapts frequency resolution to species-specific vocalizations.
- **Multi-Task Learning:** Joint training for species classification + call type detection (e.g., song vs. alarm) may improve feature representations through shared inductive biases.
- **Pruning & Sparse Networks:** Structured pruning to remove redundant channels could compress Model 1e to <350 KB while maintaining 95%+ accuracy, enabling deployment on even more constrained devices (Cortex-M4).

## References

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